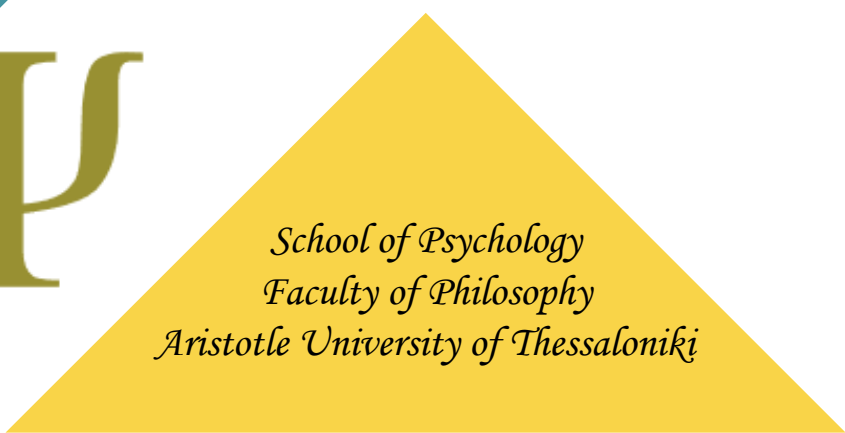




Non-parametric statistical tests

Two groups

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Έλεγχος υποθέσεων-Βήματα

1. Καθορίζεται η **μηδενική υπόθεση** H_0 (=) και **εναλλακτική υπόθεση** H_1 (≠) .
2. Ορίζεται το **επίπεδο σημαντικότητας** α (συνήθως $\alpha=0.05$).
3. Επιλέγεται μια κατάλληλη **στατιστική δοκιμασία** και υπολογίζεται η τιμή του στατιστικού με βάση τα δεδομένα του δείγματος.
4. Σύγκριση της **πιθανότητας** p να έχουμε την συγκεκριμένη τιμή του στατιστικού (ή κάτι πιο ακραίο) θεωρώντας ότι ισχύει η H_0 , με το **επίπεδο σημαντικότητας** α (0.05). Στατιστικά σημαντικό αποτέλεσμα ($p < 0.05$).
5. **Ερμηνεία** αποτελεσμάτων.

Wilcoxon-Mann-Whitney (WMW) test

WMW: Y = numeric variable → Dependent
 X = categorical (factor) with 2 levels → Independent

EXAMPLE

A high school is pilot-testing a new curriculum for mathematics. To ensure the curriculum is equitable, the department head has collected the final exam scores from a small sample of students to test for significant differences in performance between male and female students.

Y : score (maths)

X : sex (8 males, 9 females)

Hypotheses

H_0 : The medians of the two groups are equal ($\mu_{d1} = \mu_{d2}$)

H_1 : The medians of the two groups are unequal ($\mu_{d1} \neq \mu_{d2}$)



DATA

	males	rank	females	rank
	18	1	24	4
	32	10	35	12
	39	15	41	16
	26	5	34	11
	48	17	38	14
	20	2	31	9
	29	7	22	3
	30	8	27	6
			36	13
Sum of Ranks		$R_m = 65$		$R_f = 88$

$$W_m = R_m - \frac{n_m(n_m + 1)}{2} = 65 - \frac{8(8 + 1)}{2} = 29$$

$$W_f = R_f - \frac{n_f(n_f + 1)}{2} = 88 - \frac{9(9 + 1)}{2} = 43$$

In many textbooks, the statistic is the smaller W, but R specifically calculates the W for the first group listed (here females).

Wilcoxon Signed-Rank test

Wilcoxon Signed-Rank Y = numeric variable → Dependent
 X = categorical (factor) with 2 levels → Independent

Each subject is exposed to both levels of the independent variable

EXAMPLE

Researchers aim to determine whether a specific breathing technique, Box Breathing, can reduce individuals' heart rate, measured in beats per minute (BPM), immediately before they deliver a public speech. They measure the heart rates of 10 participants during a baseline stress period and again after the participants practice the breathing technique for five minutes.

Y: Beats per minute (BPM)

X: intervention (baseline measure, box breathing measure)

Hypotheses

H_0 : The median difference is zero ($\delta_{md} = 0$)

H_1 : The median difference is not zero ($\delta_{md} \neq 0$)



DATA

	Box_breath	baseline	dif	Abs dif	Rank	Sign	Signed Rank
	82	95	-13	13	9	-	-9
	84	88	-4	4	3	-	-3
	93	102	-9	9	7	-	-7
	92	91	1	1	1	+	+1
	74	85	-11	11	8	-	-8
	83	98	-15	15	10	-	-10
	112	110	2	2	2	+	+2
	83	89	-6	6	4	-	-4
	86	94	-8	8	6	-	-6
	98	105	-7	7	5	-	-5
Sum of Ranks							V+ = 3 V- = 52

In many textbooks, the statistic is the smaller V, but R specifically calculates the V for the positive ranks (here V+ = 3)